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By Swetha J

AI-Powered Company-Specific Mock Interview System

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***Abstract* - This project presents a customized AI-driven mock interview and candidate evaluation system to facilitate impartial, data-informed hiring choices. The system employs sophisticated Large Language Models (LLMs) to create interview questions tailored to specific companies and roles, mimicking real-life situations and guaranteeing relevance for every position. Responses from candidates are assessed instantly through natural language processing (NLP), sentiment analysis, and predictive analytics to gauge communication abilities, subject matter expertise, problem-solving skills, and behavioral characteristics. Automated evaluation, comprehensive performance analysis, and fit suggestions facilitate equitable and impartial candidate assessments. Furthermore, the system simplifies the complete interview process, covering question personalization, answer assessment, data analysis, and feedback creation. This AI-driven framework improves recruitment practices by enhancing reliability, allowing organizations to effectively identify the most suitable talent through reduced manual effort, decreased subjectivity, and increased consistency, accuracy, and efficiency.**

***Keywords* - AI-driven hiring, LLM-based interview questions, automated candidate evaluation, predictive hiring analytics, talent acquisition, interview management system, unbiased hiring decisions.**

I. INTRODUCTION

Recruitment plays a vital role in organizational success, as it directly impacts productivity, growth, and operational efficiency. Conventional hiring practices, including manual interviews, resume screening, and written examinations, often suffer from limitations such as inefficiency, lack of consistency, and subjective judgment. Evaluating a large pool of candidates through manual processes is time-consuming and susceptible to human biases, which can lead to less effective hiring decisions. As the demand for skilled professionals continues to rise, there is a growing necessity for intelligent recruitment systems capable of delivering efficient, unbiased, and data-driven candidate assessments.

Artificial Intelligence (AI) and Machine Learning (ML) present effective solutions for modernizing recruitment processes. By automating candidate assessments and analyzing large volumes of data systematically, these technologies can reduce bias and improve decision-making. Natural Language Processing (NLP) enables understanding of candidate responses, while predictive analytics evaluates skills and suitability in an objective manner. The advent of Large Language Models (LLMs) further enhances recruitment by generating context-aware, role-specific interview questions that simulate real-world scenarios. Such adaptive systems can customize interviews for each candidate, ensuring thorough evaluation across technical, behavioral, and problem-solving abilities.

Current AI-based recruitment tools, however, are often limited in scope. Some focus solely on resume screening or keyword matching, neglecting soft skills and personality traits. Others rely on pre-set question banks, which may not adapt to individual candidate profiles, reducing assessment relevance. Additionally, incomplete evaluation methods can allow bias to persist. Addressing these limitations requires a comprehensive system that not only automates recruitment workflows but also provides real-time insights, predictive recommendations, and fair candidate scoring.

The proposed AI-driven interview system aims to transform the hiring process by generating competency-based, role-specific questions using LLMs and analyzing responses in real time with NLP, sentiment analysis, and predictive techniques. It evaluates communication skills, technical knowledge, problem-solving ability, and behavioral characteristics, producing automated scores and detailed performance insights. This reduces manual effort, ensures consistent and unbiased evaluation, and provides actionable feedback for HR teams.

Furthermore, the system optimizes the entire recruitment workflow, from question personalization and interview conduction to response assessment and feedback generation. By combining intelligent evaluation with automation, it enhances efficiency, accuracy, and consistency, minimizes human bias, and improves the candidate experience. Overall, this AI-driven interview system provides a scalable, adaptable, and unbiased approach for modern talent acquisition.

II. LITERATURE REVIEW

B. A. Mandalapu, T. N. Kumar, M. N. Sathwika, P. Pavansai, and B. Khasim [1] introduced an AI-enabled mock interview framework that integrates speech-based confidence estimation, facial emotion analysis, and automated response assessment. The framework provides real-time feedback on confidence, emotional state, and overall interview performance to emulate realistic interview settings. By jointly processing audio and visual information, it improves interaction richness and evaluation depth. However, the system performs optimally under controlled conditions, with variations in illumination, ambient noise, and speaker accents negatively impacting accuracy.

P. V. Chintalapati, S. S. Paluri, S. S. Nikhitha, T. S. V. Nanditha, S. T. V. L. Daneswari, and P. Kiran Sree [2] presented an automated interview evaluation model that analyzes facial expressions, body gestures, vocal attributes, and NLP-based textual responses. The platform simulates real interview environments and generates customized feedback to reduce preparation effort and cost. Experimental results indicate improved training effectiveness compared to conventional mock interviews. Nevertheless, the heavy reliance on vision-based inputs constrains robustness and limits scalability in dynamic real-world scenarios.

P. Kalaiarasi, S. S. Vali, J. Sony, B. P. K. Reddy, and B. S. Reddy [3] formulated an AI- and NLP-driven interview preparation platform that combines aptitude assessment, logical reasoning, coding practice, and mock interview modules. Generative AI is employed to adaptively generate questions based on candidate responses, while speech analysis and facial emotion recognition enable personalized feedback. Although the system enhances learning outcomes and interview readiness, its dependence on cloud infrastructure raises concerns related to data privacy, operational cost, and accessibility in resource-limited environments.

M. Kaviya, R. Monisha, S. Pavithra, and R. Priyadharshini [4] engineered a multimodal mock interview evaluator that integrates facial emotion detection, audio-based confidence analysis, and NLP-driven response interpretation. Deep learning architectures and acoustic feature extraction support real-time evaluation of verbal and non-verbal behavior, improving realism and assessment accuracy. Despite these advantages, generalization across diverse datasets remains challenging, and efficient real-time multimodal fusion continues to be an open research issue.

S. K. Senthilkumar, S. Ranjith, M. Sivasakthi, and R. Ramvignesh [5] implemented an AI-driven mock interview system that combines NLP, computer vision, and reinforcement learning to dynamically regulate interview progression. The framework adapts question difficulty based on candidate performance and evaluates posture and facial cues to generate tailored feedback. While this adaptability enhances engagement, accent variability poses challenges for speech analysis, and high computational requirements limit scalability and real-time deployment.

M. K. Mythili, M. Krithika, A. Jamuna, and S. Indhulega [6] developed an AI-based mock interview application using ensemble learning and semantic text processing. The system provides real-time feedback, interactive scoring, and visual analytics to support performance improvement. By reducing evaluator subjectivity, it promotes fairness and consistency in assessment. However, accurately identifying subtle emotional variations remains a persistent limitation.

S. Rajbhar, S. Shelke, A. Singh, Y. Singh, and P. Shinde [7] devised AIced-Prep, an intelligent mock interview evaluation platform that integrates large language models, retrieval-augmented question generation, resume classification, facial emotion recognition, and speech-to-text processing. The platform offers personalized real-time feedback and standardized evaluation metrics to ensure consistent assessment. Automated summaries assist candidates in identifying skill gaps, while future enhancements target improved multilingual support and emotion recognition precision.

P. K. Mishra, A.K. Arulappan, et al. [8] outlined an AI-based virtual mock interview framework leveraging GPT-4, NLP, and machine learning to simulate realistic interview interactions. Candidate responses are evaluated through voice-based communication, and customized feedback is generated to improve preparedness. The framework significantly reduces preparation costs and reliance on human evaluators. Nonetheless, emotion-aware assessment and organization-specific customization remain areas for further development.

T. Sharma, G. Gupta, A. Singh, S. Singh, et al. [9] assembled an AI-powered interview simulation platform utilizing computer vision, NLP, and generative AI techniques. The system evaluates speech characteristics, facial expressions, and emotional indicators to assess both verbal and non-verbal communication skills. Personalized feedback enhances confidence and engagement, although system performance is highly sensitive to input quality and environmental conditions.

Y.-C. Chou, F. W. G. Wang and C.-Y. Chao [10] designed a video-centric AI mock interview platform that analyzes candidate responses using textual, vocal, and facial features. The system provides immediate feedback on emotional state, personality traits, and communication effectiveness, supporting candidate self-improvement and recruiter decision-making. However, the lack of eye-gaze tracking and multilingual functionality limits its wider applicability.

S. Khallelullah, V. S. Kumar, M. Mahati, G. Soni, and C. B. Sai [11] investigated an AI-based mock interview evaluation system that assesses verbal and non-verbal communication through facial expression analysis and speech transcription. The approach offers more objective evaluation than manual methods and reduces human bias. Its accuracy, however, declines in poor lighting and noisy environments, and high processing demands challenge real-time implementation.

N. Amarasena et al. [12] articulated an AI-enabled mock interview platform employing CNN-based multimodal analysis for confidence assessment along with LLM-driven question generation. The system delivers real-time, data-driven feedback to enhance employability outcomes. While multimodal integration enriches assessment, dependable performance requires high-quality inputs, and computational overhead constrains scalability.

A. K. Singh et al. [13] structured an AI-driven mock interview system that evaluates technical and soft skills using NLP, CNN-based facial emotion recognition, and MediaPipe-based posture analysis. The framework provides real-time feedback on communication skills, confidence, and body language, supporting holistic preparation. Nevertheless, dataset bias and computational complexity hinder large-scale deployment, particularly in low-resource settings.

S. Agarwal, J. Gupta, A. Kulshrestha, and A. Singhal [14] established an AI-based interview evaluation system capable of jointly assessing technical proficiency and emotional state. Real-time personalized feedback reduces candidate anxiety and improves readiness. Although experimental results indicate strong accuracy, further validation across diverse domains is necessary.

G. Saraf, R. Revandikar, P. Shriramwar, S. Thorat, and S. A. Jakhete [15] operationalized IamReadyAI, a real-time AI-powered mock interview platform based on speech recognition and large language models. The system automates interview simulations and supports scalable usage, reporting approximately 84% effectiveness. While automated feedback enables independent learning, limited emotion analysis and reliance on LLM-generated responses restrict evaluation depth.

S. Todsam, A. Valte, S. Kerkar, and K. Masal [16] implemented a machine-learning-based mock interview evaluation framework that combines NLP-driven response analysis with CNN-based facial emotion recognition. The platform provides immediate, personalized feedback by assessing both verbal and non-verbal cues. However, subtle emotional distinctions are difficult to capture, and performance may decline in resource-constrained environments.

T. Agarwal et al. [17] conceptualized OpportuNest, an AI-driven campus career management ecosystem developed using the MERN stack. The platform integrates resume parsing, placement prediction, mock assessments, and GPT-based interview simulations. Personalized dashboards support students, placement officers, and administrators, improving engagement and decision-making. Nonetheless, cloud dependency and scalability concerns may affect large-scale institutional adoption.

S. Nagdeote, N. Joshi, A. Serrao, S. Chiwande, and P. Dsouza [18] assembled PrepGenius, an AI-assisted job preparation platform that integrates ATS-based resume evaluation with AI-driven mock interviews. Automated verbal and non-verbal feedback enhances interview preparation, while standardized assessment ensures consistency. Limitations include single-language support and reduced adaptability due to predefined question sets.

S. Sivaramakrishnan, F. Z. Minni, A. Anand, A. Sahoo, and B. Hemang [19] constructed a CNN-based automated mock interview framework integrating facial analysis, speech processing, gesture recognition, and NLP. The system delivers real-time personalized feedback to strengthen confidence and skills. Despite improved objectivity over traditional methods, high computational demands restrict scalability and real-time efficiency.

A. Jain, S. Tibrewal, S. Jain, and G. Singh [20] presented SKILLDRILL, a machine-learning-based platform designed to simulate real-time technical interviews. The system incorporates audio-visual assessment, collaborative coding tools, and digital whiteboards to evaluate problem-solving ability. Behavioral monitoring ensures assessment integrity, while detailed reports facilitate targeted skill improvement. Overall, the platform provides an efficient and scalable solution for technical interview preparation.

III. Methodology

A. Candidate Data Collection and Feature Extraction

This module handles the collection of candidate resumes and profile information as the initial input to the system. It applies resume parsing and preprocessing methods to identify and extract essential attributes, including academic qualifications, technical skill sets, work experience, and job-specific competencies. The processed data is then organized into a structured format, enabling the system to use it as contextual input for generating adaptive interview questions and conducting personalized candidate evaluations.

B. LLM-Based Adaptive Interview Question Generation

In this module, Large Language Models (LLMs) are employed to dynamically generate context-aware and role-specific interview questions. The question generation process leverages extracted candidate features and predefined job requirements to assess both technical knowledge and behavioural attributes. This adaptive mechanism ensures consistency across interviews while maintaining personalization for each candidate.

C. AI-Driven Interview Execution and Response Evaluation

Candidates take part in a structured, multi-round interview process delivered through an AI-enabled virtual interview platform. Their responses are evaluated instantaneously through NLP-based and machine learning–driven analysis to assess semantic relevance, clarity of communication, sentiment, and problem-solving ability. This automated evaluation reduces human subjectivity and promotes unbiased assessment throughout all stages of the interview process.

D. Automated Performance Analysis and Hiring Decision Support

This module aggregates results from resume screening and interview evaluations to generate automated scores and detailed performance analytics. The system produces comprehensive assessment reports and candidate rankings, enabling recruiters to make informed, data-driven, and fair hiring decisions. By reducing manual intervention, this module enhances recruitment efficiency and overall decision accuracy.

Overall, the integration of these modules forms a cohesive and intelligent recruitment framework that supports end-to-end candidate evaluation. By combining resume-based feature extraction, LLM-driven interview personalization, real-time response analysis, and automated performance analytics, the system ensures consistency, scalability, and fairness throughout the hiring process. This modular architecture enables seamless information flow between components while reducing human bias and manual effort, thereby improving the accuracy and efficiency of hiring decisions across diverse job roles and organizational contexts.

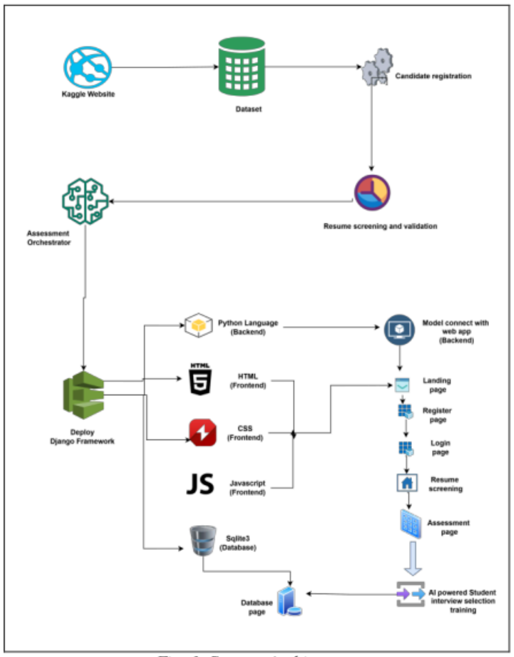

Fig. 1. System Architecture

IV. Results and Analysis

An AI-enabled interview and candidate assessment framework was evaluated to determine its effectiveness in automating recruitment activities while ensuring consistency in candidate evaluations. The system generated adaptive, role-oriented interview questions using advanced computational language models, enabling the interview process to align with individual candidate profiles and job-specific requirements. The structured, multi-level interview design supported a comprehensive assessment of candidates across technical competencies, communication abilities, and behavioral characteristics.

The response analysis component demonstrated reliable real-time evaluation by applying NLP and ML techniques to interpret candidate answers. Key aspects such as contextual relevance, clarity of expression, and sentiment were assessed automatically, leading to uniform scoring outcomes across interviews. This approach minimized dependence on human judgment and significantly reduced evaluation bias, thereby improving the objectivity of the assessment process.

From an operational standpoint, the integrated framework enhanced recruitment efficiency by consolidating resume screening, interview execution, and performance reporting into a single workflow. Recruiters received structured insights and candidate rankings that supported quicker and more informed selection decisions. Overall, the results confirm that the system improves evaluation accuracy, scalability, and fairness, making it well suited for contemporary data-driven recruitment environments.

TABLE 1: Performance Metrics

Metric	Value(%)
Accuracy	94.5
Precision	92.8
Recall	93.2
F1-Score	92.5

V. Conclusion and Future Works

This study presents an intelligent interview and candidate assessment framework designed to improve the reliability, efficiency, and fairness of hiring practices. The proposed system combines advanced language-understanding models and data-driven learning techniques to support dynamic question generation, instant evaluation of candidate responses, and automated scoring. By reducing reliance on manual review and subjective judgment, the framework promotes consistency in candidate evaluation and supports evidence-based recruitment decisions. Its modular structure and performance outcomes demonstrate its ability to streamline hiring workflows while maintaining evaluation precision.

Future work will focus on extending the assessment approach through multimodal capabilities, including the analysis of speech patterns, facial behavior, and other non-verbal indicators to enhance candidate profiling. Additional improvements may involve adopting domain-focused intelligent models, incorporating interpretable AI methods to improve transparency in decision-making, and enabling large-scale deployment supported by continuous learning and bias-mitigation strategies. These developments aim to enhance the system's flexibility, clarity, and practical applicability across diverse recruitment settings.

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